

Introduction

Ordinary differential equation (ODE) models are central to quantitative biology, pharmacology and epidemiology, where they describe how interacting molecular species or populations change over time as a function of kinetic parameters θ [1]. Using a model for prediction requires estimating θ from experimental measurements, and this becomes difficult as models grow. The main obstacle is non-identifiability: different parameter values may produce the same measured output, either because of model symmetries (structural non-identifiability) or because the available data are insufficient (practical non-identifiability)[2–4]. In both cases the likelihood has flat ridges on which optimisers stall and confidence intervals become unreliable[5–7]. Models in public benchmark collections routinely contain 20–300 parameters[8], and genome-scale models contain thousands[9].

The sloppy-model phenomenon offers a structural explanation[10]: the sensitivity of model outputs to parameters is typically dominated by a few “stiff” directions, while many “sloppy” directions have almost no effect on the outputs. This suggests that a small subset of parameters may suffice to describe what is actually measured. Exploiting this observation in practice requires answering three questions that existing tools address only partially. First, which parameters should be retained? Sensitivity indices[11, 12] and profile likelihoods[5] rank parameters by importance, but they do not ensure that the retained parameters are mutually non-redundant, so the reduced estimation problem may still be ill-conditioned[13, 14]. Second, is the reduced model good enough? A reduction is useful only if re-estimating the retained parameters reproduces the measured data and predicts experiments that were not used for fitting. Third, how much can the conclusions be trusted? Statements such as “the discarded parameters can be fixed” rest on approximations whose validity is rarely made explicit.

This work was motivated by a dynamical model of Bandura’s social cognitive theory[15], in which six psychosocial constructs interact through 22 parameters (coupling strengths, input gains and time constants). Attempts to estimate all of them showed that several parameters have practically identical effects on the model output, so that no amount of data can separate them; in such models, a principled way to decide which parameters to estimate is indispensable.

Here we present a reduction method that works directly on the measured outputs defined in the PETab format[16], including experimental conditions, measurement transformations and noise levels. It distinguishes biological parameters, which belong to the mechanistic model, from calibration parameters (scaling factors and offsets of the measurement process), which are always re-estimated, and selects only among the biological ones. Candidates are ranked by sensitivity, redundant candidates are rejected by condition-number and variance-inflation tests, the reduced model is accepted only if re-estimating its parameters reproduces the full model under random parameter perturbations, and a final backward elimination makes the subset minimal with respect to inclusion. We evaluate the method on 34 benchmark problems with real data[8, 16], test the reduced models on held-out experiments, quantify their robustness to perturbation size and to the method’s own settings, and report the computational cost. The main mathematical statements on which the method relies are

formally verified in the Lean 4 proof assistant with the Mathlib library[17, 18]; we state precisely what is and what is not proved.

Results

Overview of the method

Figure 1 summarises the procedure; all symbols are defined in the Methods and in Supplementary Table S2. The input is a PETab problem: an SBML model, the list of measured observables with their experimental conditions, measurement times, transformations (linear, logarithmic) and noise standard deviations σ_i , and the nominal parameter values θ_0 reported by the original authors. Each measurement i defines a scaled output $y_i(\theta) = h(g_i(x(t_i; \theta), \theta))/\sigma_i$, so that all outputs are comparable in units of measurement noise. Parameters are split into biological parameters θ_B (those of the SBML model) and calibration parameters θ_C (observation scalings and offsets); only biological parameters are candidates for selection, while calibration parameters are always re-estimated together with the selected ones.

The relative sensitivity matrix $J = \partial y / \partial \log \theta$ is computed once at θ_0 by central finite differences. In stage 1, biological parameters are added in decreasing order of their sensitivity energy $E_j = \|J_{:,j}\|^2$; a candidate is accepted only if the condition number κ and the largest variance inflation factor (VIF) of the normalised columns of J_S stay below a threshold, and is otherwise discarded permanently. Stage 1 stops when the accepted parameters capture a fraction τ_R of the total energy. The resulting subset S is then tested: in random scenarios where all parameters are perturbed, θ_S and θ_C are re-estimated with the remaining parameters fixed at θ_0 , and S is admissible if the median relative fitting error e_{fit} is below τ_e and $|S| \geq 2$. The threshold $\tau_e = \sqrt{1 - c^2}$ equals the smallest error attainable by rescaling an output change whose cosine similarity with the full change is c (Theorem 1); we use $c = 0.9$, so $\tau_e = 0.436$. If stage 1 does not yield an admissible subset, stage 2 repeats the selection with parameters ordered by their finite-perturbation effect s_j until an admissible subset is found. Finally, backward elimination removes any parameter whose removal keeps the subset admissible, which yields a subset that is minimal with respect to inclusion (Theorem 2).

A motivating case with exact collinearity

The social cognitive theory model[15] is a linear system of six constructs, $\tau_i \dot{x}_i = -x_i + \sum_j \beta_{ij} x_j + \sum_k \gamma_{ik} u_k$, with 22 parameters, three input conditions and the fourth construct as measured output (Methods). It has no experimental data or calibration parameters, so it is analysed with simulated outputs and is not counted in the benchmark below. The sensitivity matrix of all 22 parameters is exactly rank-deficient: its condition number and largest VIF are infinite, and 55 pairs of parameters have sensitivity vectors with cosine similarity of at least 0.95, several of them equal to 1.000 (for example β_{43} and γ_{33} , β_{46} and γ_{64} , or τ_1 and τ_5). Estimating all parameters is therefore impossible. The method selected two parameters, β_{43} and τ_4 ; the conditioning test rejected 15 candidates, including γ_{33} , whose effect on the output is identical to that

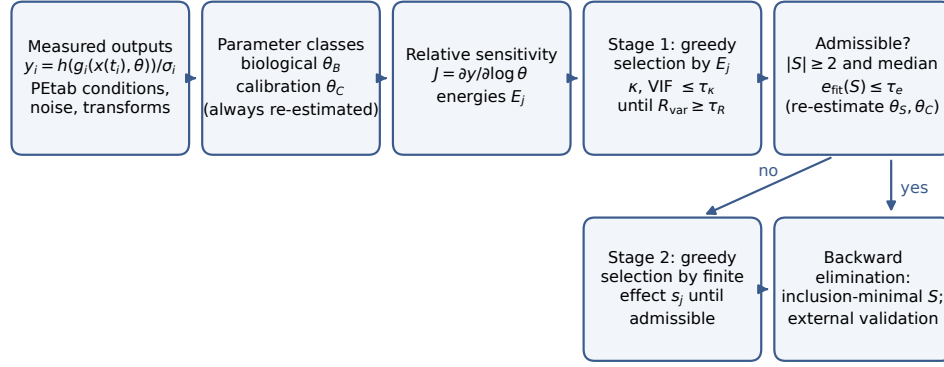


Fig. 1 Overview of the reduction method. Measured outputs are scaled by their noise level; biological and calibration parameters are separated; the relative sensitivity matrix J is computed at the nominal parameters. Stage 1 selects biological parameters in decreasing order of sensitivity energy, rejecting candidates that make the selection ill-conditioned (κ or VIF above τ_κ), until a fraction τ_R of the energy is captured. The subset is admissible if re-estimating it, together with the calibration parameters, reproduces the full model in random perturbation scenarios (median $e_{\text{fit}} \leq \tau_e$) and it has at least two parameters. Otherwise stage 2 repeats the selection ordered by finite-perturbation effects. Backward elimination then removes unnecessary parameters, and the reduced model is validated on held-out experiments.

of β_{43} , and τ_3 (cosine 0.988 with τ_4). Re-estimating only these two parameters reproduced the output with a median e_{fit} of 2.0% at $\pm 5\%$, and the error stayed between 1.5% and 2.1% for perturbations up to $\pm 50\%$ (pattern A). When all six constructs are treated as outputs, the redundancy is smaller (9 pairs with cosine ≥ 0.95 , still with an infinite condition number) and the method selects γ_{33} and β_{43} ($e_{\text{fit}} = 17.7\%$, pattern A).

Benchmark: a few biological parameters suffice in most models

We applied the method to 34 PEtAb benchmark problems with real measurements [8, 16] covering signalling, metabolism, gene regulation, cell cycle, immunology and epidemiology, with 3 to 294 parameters and 16 to 27,132 measurements (Table 1). One further benchmark problem (Froehlich, 4,231 parameters and 9,169 experimental conditions) was analysed only on its state trajectories because each evaluation of its measured outputs requires about 9,000 simulations; it is not counted here (see Computational cost).

The method found an admissible subset in 32 of the 34 problems (Fig. 2 and Table 1). In 29 of them the subset contains only two biological parameters; the remaining three contain three (Zhao), five (Bruno) and ten (Lucarelli). Sixteen problems also have calibration parameters (up to 79 in Raimundez), which are re-estimated in every case. Counting both kinds, the reduced estimation problem has a median of 78% fewer free parameters than the full one (range 33–97%); among biological parameters alone, the median reduction is 88%. Backward elimination removed between one and five parameters in seven problems (Supplementary Table S1). Stage 1 alone was sufficient

in seven problems; in the other 25, the energy ordering selected parameters whose re-estimation could not reproduce the full model, and the finite-effect ordering of stage 2 was needed. The two non-admissible problems fail for different reasons. In Crauste, a model of the T-cell response with quadratic proliferation and pathogen-growth terms, the published parameters lie at the edge of a regime in which the solution grows without bound in finite time: changing a single parameter by 1% (for example, decreasing the effector proliferation rate ρ_E by 1%) makes the effector populations increase from about 5×10^2 to 5×10^5 cells within four days, after which no solution exists on the measured horizon. This happened for 8–9 of the 24 one-at-a-time perturbations of $\pm 1\%$ and for 7 of the 15 scenarios at $\pm 5\%$, independently of the integrator tolerances (relative tolerances from 10^{-10} to 10^{-5}), so the sensitivity matrix and the admissibility test are not defined and no local reduction method is applicable; consistently, global existence of solutions for this model could only be proved formally on a short horizon ($T \leq 1$). In Beer, the selection stopped at 18 parameters because any further parameter violated the conditioning test, and the resulting subset had $e_{\text{fit}} = 50\%$, above the threshold.

Re-estimation is essential. When the selected parameters are simply set to their perturbed values, without re-estimation, the median relative error e_{rel} exceeds τ_e in 23 of the 32 admissible problems (Fig. 2b); after re-estimation the median error e_{fit} is 14% (maximum 43.5%). In other words, the selected parameters do not replicate the effect of every individual perturbation; rather, adjusting them compensates for the perturbations of all other parameters in the measured outputs. This is precisely the property needed when the reduced model is fitted to data.

Reduced models predict held-out experiments

Reproducing the full model is not the same as predicting new data. We therefore split the measurements of each problem into training and test sets, by experimental condition when several conditions were available (20 problems) and by time otherwise, using the later time points as test set (11 problems); calibration or condition-specific parameters that appear only in the test set were excluded. The reduced model (re-estimating θ_S and θ_C) and, for problems with up to 60 parameters, the full model (re-estimating all parameters) were fitted to the training data starting from θ_0 and evaluated on the test data. In two problems a split was impossible (Bertozzi, whose selected parameters only act in the test conditions; Blasi, with a single condition and a single time point).

The reduced model predicted the held-out experiments as well as or better than the full model in 24 of 25 problems in which both could be compared (Fig. 3a; in two of these, Zhao and Isensee, the two test errors differed by less than 0.2%). The exception was Raia (test χ^2/n of 8.2 versus 5.1). In several problems the full model overfitted the training data and predicted much worse, for example Brannmark (test χ^2/n 5.8 versus 2,145), Weber (1.0 versus 226) and Zheng (1.6 versus 20.6). Compared with the published estimate θ_0 , which was obtained using all data including the test set, the reduced model reached a test χ^2/n no larger than twice that of θ_0 in 22 of 29 problems (Fig. 3b). In three problems (Okuonghae, Oliveira, Smith) even θ_0 has

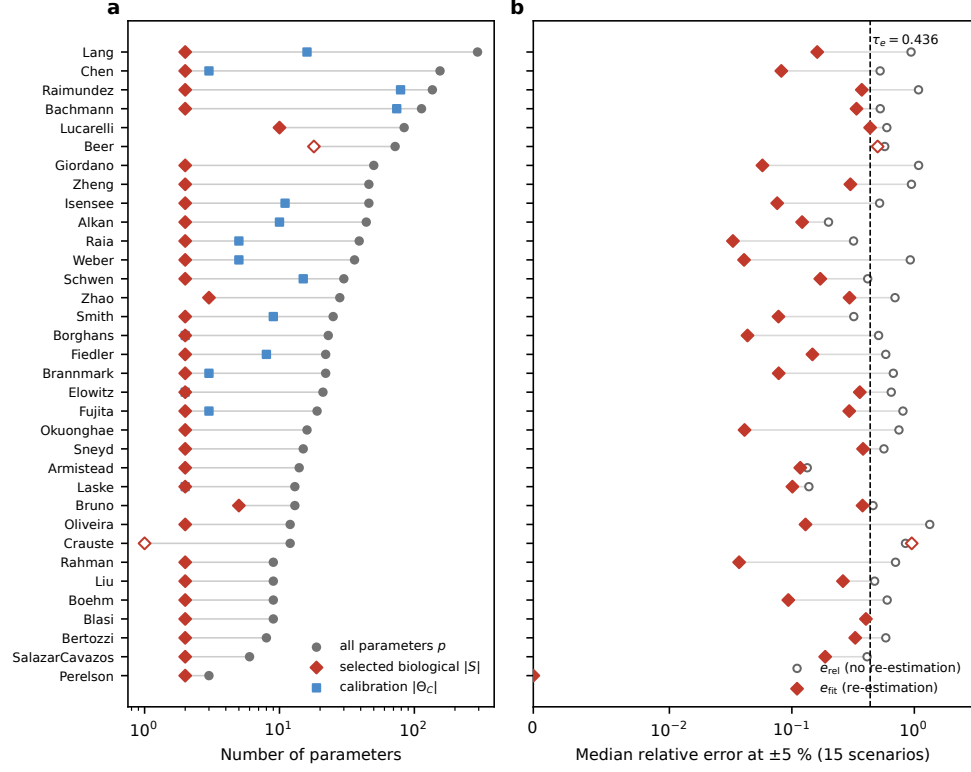


Fig. 2 Reduction across 34 benchmark problems. **a**, Number of parameters of the full model (p , grey circles), calibration parameters that are always re-estimated ($|\Theta_C|$, blue squares) and selected biological parameters ($|S|$, red diamonds; open diamonds, no admissible subset). Problems are sorted by p . **b**, Median relative error over 15 random scenarios at $\pm 5\%$ when the selected parameters are set to their perturbed values without re-estimation (e_{rel} , open circles) and after re-estimating them together with the calibration parameters (e_{fit} , red diamonds). The dashed line marks the admissibility threshold $\tau_e = 0.436$, equivalent to a cosine similarity of 0.9. The horizontal axis is linear below 0.01 and logarithmic above.

182 $\chi^2/n \gg 1$, which indicates that the noise levels recorded in the benchmark do not
 183 match the scatter of the data.

184 We also performed a synthetic experiment in which the truth is known. For each
 185 problem we drew a true parameter vector θ^* within $\pm 20\%$ of θ_0 , simulated noisy data
 186 with the benchmark noise levels, fitted the reduced model on the training conditions
 187 and measured its relative prediction error e_{pred} on the test conditions. In 15 problems
 188 the effect of the parameter change on the test outputs was smaller than the measure-
 189 ment noise (χ^2/n at θ_0 below 2), so e_{pred} is not informative; in 13 of these 15 the
 190 reduced model predicted at the noise level ($\chi^2/n \leq 2$). Among the 13 problems where
 191 the signal exceeded the noise, the reduced model achieved $e_{\text{pred}} \leq \tau_e$ in 8 and predicted
 192 as well as or better than the full model in 6 of 11 comparable cases. Thus, with real-
 193 istic noise the reduced models predict at least as well as the full models on real data,

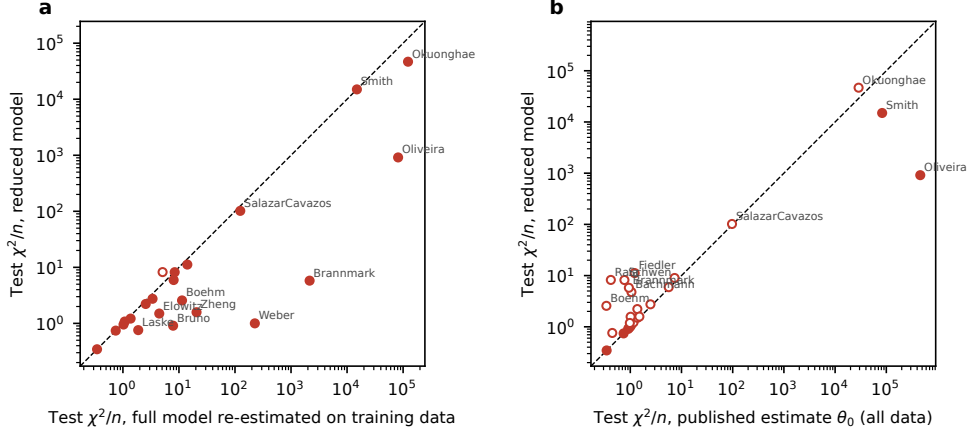


Fig. 3 Prediction of held-out experiments with real data. Mean squared noise-scaled residual (χ^2/n) on the test experiments for the reduced model (vertical axis) against **a**, the full model re-estimated on the same training experiments ($n = 25$ problems with $p \leq 60$) and **b**, the published estimate θ_0 , which was obtained using all data including the test experiments ($n = 29$). Filled symbols: reduced model within 5% of, or better than, the reference; open symbols: worse. Points below the dashed diagonal mean that the reduced model predicts better. Problems with noise-free benchmark data (test χ^2/n of θ_0 below 0.01; Giordano, Perelson, Rahman) are omitted. Names are shown for points away from the diagonal.

194 while the synthetic experiment shows that the prediction accuracy of a two-parameter
 195 model has limits when parameter changes are large relative to the noise.

196 Robustness to the size of parameter changes

197 Admissibility is decided with perturbations of $\pm 5\%$. To determine how far the con-
 198 clusion extends, we kept each final subset fixed and repeated the evaluation with
 199 perturbations of $\pm 1, 5, 10, 20, 30, 40$ and 50% (15 scenarios per level, same random
 200 seed at every level, so that the $\pm 5\%$ level reproduces the selection). We classified the
 201 behaviour of the median e_{fit} into four patterns: A, admissible at all levels; B, admis-
 202 sible up to at least $\pm 20\%$ and not thereafter; C, irregular (failing and then passing
 203 again); D, failing at $\pm 10\%$ or less.

204 All 32 admissible problems were evaluated. Twenty-seven follow pattern A (Fig. 4):
 205 their median e_{fit} stays below the threshold up to $\pm 50\%$, typically varying by only a
 206 few percentage points across levels. Four follow pattern B: Armistead, whose error
 207 grows steadily from 3% at $\pm 1\%$ to 63% at $\pm 50\%$, and Bertozzi, Blasi and Elowitz,
 208 which are close to the threshold already at $\pm 5\%$ and cross it between $\pm 20\%$ and
 209 $\pm 30\%$. Lucarelli is the only case of pattern C: its median error stays between 43%
 210 and 47% at all levels, that is, on the threshold rather than clearly above or below it.
 211 No problem follows pattern D. The reductions therefore remain valid well beyond the
 212 $\pm 5\%$ used for selection in most problems, whereas problems that are admissible only
 213 with a small margin at $\pm 5\%$ are those that lose admissibility for large changes.

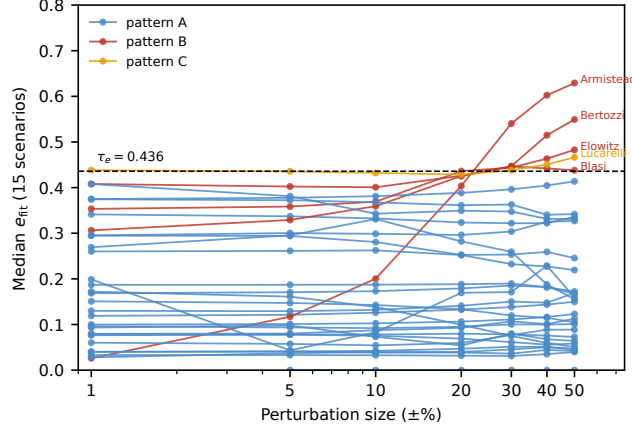


Fig. 4 Robustness of the selected subsets to the size of parameter changes. Median relative error after re-estimation (e_{fit} , 15 random scenarios per level) as a function of the perturbation size, for the final subset of each of the 32 admissible problems. Colours indicate the robustness pattern: A, admissible at all levels (blue, $n = 27$); B, admissible up to at least $\pm 20\%$ (red, $n = 4$); C, irregular (orange, $n = 1$). The dashed line marks the admissibility threshold $\tau_e = 0.436$.

Sensitivity to method settings and computational cost

The method depends on a finite-difference step δ , integrator tolerances, the perturbation size used for admissibility and four thresholds (τ_R , τ_κ , τ_{VIF} , τ_e). We re-ran the complete method on the 26 small and medium problems changing one setting at a time (Table 2). The selected subset was identical to the baseline in all 26 problems for $\delta = 0.1\%$ and 5% , for perturbations of $\pm 1\%$ and $\pm 10\%$, and for τ_R between 0.80 and 0.95, and in 25 of 26 with tighter integrator tolerances; admissibility decisions changed in at most one problem (Perelson at $\pm 1\%$, where the perturbed output change falls below the numerical noise floor). The κ/VIF threshold changed the subset in three or four problems without changing the number of admissible problems. As expected, the error threshold τ_e has the largest influence: requiring $e_{\text{fit}} \leq 10\%$ leaves 22 admissible problems with a mean of 3.6 biological parameters, whereas τ_e between 0.30 and 0.50 gives 25 admissible problems with 2.1–2.4 parameters. The threshold therefore acts as an accuracy–parsimony trade-off that the user can set; we also examined single-parameter subsets: in ten problems one parameter reached the threshold at $\pm 5\%$, but in the synthetic prediction experiment the selected pair predicted better in six of nine comparable problems, which supports the minimum size of two.

The dominant cost is the computation of J ($2p$ simulations) and of the stage-2 ordering (p simulations), so the number of simulations grows linearly with the number of parameters, and all of them are independent and can run in parallel; the selection, condition-number tests, backward elimination and the first-order version of e_{fit} only use linear algebra on J . On a four-core server, these simulations took less than 40 s for all but two of the problems we timed, 2 min for Lang (294 parameters) and 68 min for Chen (155 parameters, 35 s per simulation). On state trajectories, the 4,231-parameter Froehlich model required 12,693 simulations and 2.6 h. For models of this size on

measured outputs, forward or adjoint sensitivity equations[19] compute J at the cost of a few simulations.

Formal guarantees

Table 3 lists the mathematical statements on which the method relies and that are verified in Lean 4 with Mathlib; Supplementary Table S3 gives the name of each formal theorem. None of the proofs uses unproved assumptions (`sorry`) or additional axioms. Two statements are central to the method:

Theorem 1. *For vectors $d \neq 0$ and $b \neq 0$ in a real inner-product space, $\min_{\lambda \in \mathbb{R}} \|d - \lambda b\|/\|d\| = \sqrt{1 - \cos^2(d, b)}$.*

Theorem 2. *Let $\mathcal{A}$ be any admissibility predicate on finite sets and S an admissible set. Backward elimination (removing one element at a time whenever the remaining set is still admissible, until no element can be removed) returns an admissible subset of S from which no element can be removed without losing admissibility.*

Theorem 1 justifies the threshold τ_e : an error of at most $\sqrt{1 - c^2}$ after re-estimation is the natural counterpart of a cosine similarity of at least c . Theorem 2 gives the precise sense in which the selected subsets are minimal: no parameter can be removed, although a different, smaller admissible subset that is not contained in the result may exist. The other verified statements cover the remaining steps: re-estimating more parameters never increases the fitting error and $e_{\text{fit}} \leq e_{\text{rel}}$; the selection keeps κ and VIF below their thresholds and permanently discarding a rejected candidate is correct because both quantities can only increase when parameters are added; κ and VIF bounds imply unique and stable least-squares estimates; error bounds on the states transfer to the measured outputs, including logarithmic transformations; and the first-order version of e_{fit} used for the largest models differs from the exact one by a second-order term. The verification is local: it does not prove that a subset remains adequate far from θ_0 , which is assessed empirically above.

Discussion

The central finding is that, for most published systems biology models with real data, the measured outputs can be reproduced by re-estimating only two biological parameters plus the measurement calibration parameters, and that such reduced models predict held-out experiments as well as or better than the full models. This is consistent with the sloppy-model picture[10], but it is a stronger and more practical statement: it refers to the outputs that were actually measured, under the actual experimental conditions and noise levels, and it is checked by re-estimation and external validation rather than inferred from the sensitivity spectrum alone. The two-stage design reflects a practical observation: ranking by local sensitivity energy (stage 1) was sufficient in only seven problems, whereas ranking by finite-perturbation effects (stage 2) was needed in most. Energy measures how strongly a parameter moves the outputs, not

whether adjusting it can compensate for changes in other parameters; the admissibility test, which re-estimates the subset, is what decides.

The method complements existing approaches. Profile likelihood[5, 6] and likelihood-based observability analysis[7] assess identifiability after calibration and require repeated optimisation in the full parameter space; Bayesian approaches such as approximate Bayesian computation[20] require many more simulations. Global sensitivity analysis[11, 12] ranks parameters but does not produce a non-redundant subset validated by re-estimation, and structural identifiability analysis[2–4] addresses ideal noise-free data rather than a given experimental design. Model-order reduction methods such as balanced truncation[21, 22] reduce the number of state variables rather than parameters. Our method requires a number of simulations that grows linearly with the number of parameters, is applied before calibration and returns a concrete subset with documented fidelity, prediction and robustness.

Several limitations should be noted. First, the analysis is local: sensitivities are computed at the published estimate θ_0 , which is assumed to be a good fit to the data, and admissibility is tested with perturbations around it. The robustness analysis shows that most reductions remain valid for changes of up to $\pm 50\%$, but this cannot be assumed for every model, and problems that are admissible with a small margin at $\pm 5\%$ (Bertozzi, Blasi, Elowitz, Lucarelli) are those that lose admissibility for larger changes. Second, the method assumes that the model is smooth in the parameters, that the measurement noise is Gaussian with the standard deviations recorded in the benchmark and that the calibration parameters can always be re-estimated; in three problems the recorded noise levels do not match the scatter of the data, and in problems with many calibration parameters (Bachmann, Raimundez) the reduced problem is still sizeable. Third, for the eight problems with the most expensive simulations the fitting error was computed to first order; the verified bounds show that the difference from the exact error is of second order in the perturbation, but it was not quantified numerically for each model. Fourth, one benchmark problem with 9,169 experimental conditions was too expensive to analyse on its measured outputs with finite differences, and two problems had no admissible subset: Beer, and Crauste, whose model has no solution on the measured horizon for parameter changes of 1%. Finally, the formal verification covers the mathematical steps of the method and the regularity of the translated models, not the numerical simulator itself; we cross-checked our simulations against the reference simulations distributed with the benchmark.

The social cognitive theory model, which motivated this work, illustrates the same behaviour outside molecular biology: exact collinearity makes the full estimation problem ill-posed, and the conditioning test identifies two parameters that carry the information in the output. In practice, the method can be used as a preprocessing step: before calibrating a large model, select the few biological parameters that the data can inform, re-estimate them together with the calibration parameters, keep the rest at literature values, and verify the reduced model on held-out experiments. The threshold τ_e provides an explicit trade-off between accuracy and parsimony. Natural extensions include computing J with forward or adjoint sensitivity equations for genome-scale models, combining sensitivities at several operating points, and using the selected subsets to guide experimental design[23].

Methods

Models, data and simulation

We used the problems of the PETab benchmark collection[8, 16] that provide real measurements, an SBML model and nominal parameter values (34 problems; Table 1). Each problem specifies the observables g_i , their transformation h (linear, log or $\log_{10}$), the noise standard deviation σ_i of each measurement, the experimental conditions (parameter or initial-value overrides) and, where applicable, pre-equilibration conditions. Models were simulated with libRoadRunner[24] (CVODE, relative and absolute tolerances 10^{-7} and 10^{-10}). Initial assignments that depend on parameters were re-evaluated for every parameter vector, condition-specific overrides were applied before simulation, pre-equilibration was run to steady state before the main simulation, and models with discontinuous inputs were integrated piecewise between discontinuities. Simulated observables were compared with the reference simulations distributed with the benchmark wherever available (Supplementary Note).

Social cognitive theory model

The model has six states x_i (constructs of social cognitive theory[15]) obeying $\dot{x} = A(\theta)x + B(\theta)u$, with $A_{ii} = -1/\tau_i$, $A_{ij} = \beta_{ij}/\tau_i$ for the ten couplings $\beta_{14}, \beta_{21}, \beta_{25}, \beta_{31}, \beta_{34}, \beta_{42}, \beta_{43}, \beta_{45}, \beta_{46}, \beta_{54}$, input gains $\gamma_{33}, \gamma_{35}, \gamma_{36}, \gamma_{57}, \gamma_{64}, \gamma_{68}$ divided by the corresponding τ_i , and six time constants $\tau_1, \dots, \tau_6$ (22 parameters). Three input conditions (unit input on the first channel, on the second channel, and both at 0.5) were simulated from $x(0) = 0$ on $[0, 10]$ with a fourth-order Runge–Kutta scheme on 80 equally spaced points. The measured output is $y = x_4$ (240 values); the variant with all states uses the 1,440 state values. All parameters are model parameters, the outputs are not scaled by noise, and all other settings are as for the benchmark problems.

Measured outputs and parameter classes

For a parameter vector $\theta \in \mathbb{R}^p$ (in the units of the benchmark, excluding noise parameters), the vector of scaled outputs is $y(\theta) \in \mathbb{R}^m$ with entries

$$y_i(\theta) = \frac{h(g_i(x(t_i; \theta, c_i), \theta))}{\sigma_i}, \quad i = 1, \dots, m, \quad (1)$$

where $x(t; \theta, c)$ is the state trajectory under condition c and (t_i, c_i) are the time and condition of measurement i . Parameters that appear in the SBML model are biological (Θ_B); parameters that appear only in the observation functions and not in the model or the condition table (scalings and offsets) are calibration parameters (Θ_C). Only biological parameters are candidates for selection; calibration parameters are always re-estimated.

356 Sensitivities, energy and conditioning

357 The relative sensitivity matrix $J \in \mathbb{R}^{m \times p}$ is computed by central differences,

$$J_{:,j} = \frac{y(\theta_0 + \delta\theta_{0,j}e_j) - y(\theta_0 - \delta\theta_{0,j}e_j)}{2\delta}, \quad (2)$$

358 which approximates $\partial y / \partial \log \theta_j$. The energy of parameter j is $E_j = \|J_{:,j}\|^2 =$
 359 $(J^\top J)_{jj}$ and the fraction of energy captured by a subset $S \subseteq \Theta_B$ is $R_{\text{var}}(S) =$
 360 $\sum_{j \in S} E_j / \sum_{j \in \Theta_B} E_j$. For a subset S , the columns of J_S are normalised to unit length,
 361 $Z = J_S \text{diag}(\|J_{:,j}\|)^{-1}$; the condition number is $\kappa(S) = \sigma_{\max}(Z) / \sigma_{\min}(Z)$ and the
 362 largest variance inflation factor is $\text{VIF}(S) = \max_j [(Z^\top Z)^{-1}]_{jj}$ [13, 14]. Normalisation
 363 removes differences of scale between parameters, so that κ and VIF measure only the
 364 angles between sensitivity directions.

365 Selection

366 *Greedy selection.* Candidates are visited in a given order. A candidate j is accepted
 367 if $\kappa(S \cup \{j\}) \leq \tau_\kappa$ and $\text{VIF}(S \cup \{j\}) \leq \tau_{\text{VIF}}$; otherwise it is discarded permanently.
 368 Because κ and VIF can only increase when columns are added, a discarded candidate
 369 would also be rejected later. *Stage 1* visits biological parameters in decreasing order of
 370 E_j and stops when $R_{\text{var}}(S) \geq \tau_R$. *Stage 2*, used if the stage-1 subset is not admissible,
 371 visits biological parameters in decreasing order of the finite-perturbation effect $s_j =$
 372 $\|y(\theta_0 + \eta\theta_{0,j}e_j) - y(\theta_0)\|$ and stops at the first admissible subset. *Backward elimination*
 373 visits the parameters of the selected subset and removes each one whose removal leaves
 374 the subset admissible.

375 Admissibility and re-estimation

376 Admissibility is evaluated on N random scenarios $\theta^{(k)} = \theta_0 \odot (1 + \varepsilon u^{(k)})$, with $u^{(k)}$
 377 uniform in $[-1, 1]^p$. For each scenario let $\Delta y_{\text{full}} = y(\theta^{(k)}) - y(\theta_0)$. The relative error
 378 without re-estimation is $e_{\text{rel}} = \|\Delta y_{\text{full}} - \Delta y_S\| / \|\Delta y_{\text{full}}\|$, where Δy_S is the output
 379 change when only the parameters in $S \cup \Theta_C$ take their scenario values, and the
 380 cosine similarity is $\cos \Delta = \langle \Delta y_{\text{full}}, \Delta y_S \rangle / (\|\Delta y_{\text{full}}\| \|\Delta y_S\|)$. The relative error after
 381 re-estimation is

$$e_{\text{fit}} = \min_{\theta_S, \theta_C} \frac{\|y(\theta_S, \theta_C, \theta_{0,\bar{S}}) - y(\theta^{(k)})\|}{\|\Delta y_{\text{full}}\|}, \quad (3)$$

382 where $\theta_{0,\bar{S}}$ denotes the nominal values of all other biological parameters. The min-
 383 imisation is carried out in $\log |\theta|$ within $[\|\theta_0\|/10, 10\|\theta_0\|]$ by a trust-region reflective
 384 least-squares method (SciPy [25], at most $100(|S| + |\Theta_C| + 1)$ function evaluations),
 385 from two starting points (the scenario value and θ_0); the error without re-estimation
 386 is also included, so that $e_{\text{fit}} \leq e_{\text{rel}}$. Scenarios whose output change is below
 387 $\max(10^{-10}\|y(\theta_0)\|, 10^{-6})$ are discarded. S is admissible if $|S| \geq k_{\min}$ and the median
 388 of e_{fit} over the scenarios is at most $\tau_e = \sqrt{1 - c^2}$. For the eight problems in which the
 389 cost of simulation made repeated re-estimation impractical (marked in Table 1), the
 390 first-order version $\min_c \|\Delta y_{\text{full}} - J_{S \cup \Theta_C} c\| / \|\Delta y_{\text{full}}\|$ was used.

Settings

Unless stated otherwise: $\delta = 0.01$; $\eta = 0.1$; $\tau_R = 0.89$; $\tau_\kappa = \tau_{\text{VIF}} = 10$; $N = 15$ scenarios; $\varepsilon = 0.05$; random seed 42; $c = 0.9$, so $\tau_e = 0.436$; $k_{\min} = 2$. The sensitivity analysis (Table 2) changed one setting at a time on the 26 problems that could be re-run completely in reasonable time.

External validation

Measurements were split into training and test sets by experimental condition (every third condition after sorting, or the second of two conditions, used for testing) or, for problems with a single condition, by time (the last third of the distinct measurement times). Parameters that only affect test conditions were excluded from re-estimation. The reduced model (re-estimating $S \cup \Theta_C$) and the full model (re-estimating all parameters; only for $p \leq 60$ because of cost) were fitted to the training data from θ_0 with the same least-squares method and bounds, and evaluated by the mean squared noise-scaled residual $\chi^2/n = \frac{1}{n} \sum_{i \in \text{test}} (y_i(\hat{\theta}) - \tilde{y}_i)^2$, where $\tilde{y}_i$ are the scaled measurements. In the synthetic experiment, the truth θ^* had each parameter drawn uniformly within $\pm 20\%$ of θ_0 , data were simulated with Gaussian noise of standard deviation σ_i , and $e_{\text{pred}} = \|y_{\text{test}}(\hat{\theta}) - y_{\text{test}}(\theta^*)\| / \|y_{\text{test}}(\theta^*) - y_{\text{test}}(\theta_0)\|$; the effect of the parameter change was considered larger than the noise when χ^2/n of θ_0 on the synthetic test data was at least 2.

Robustness patterns

For each admissible problem, the final subset was kept fixed and e_{fit} was evaluated at perturbation sizes $\varepsilon \in \{0.01, 0.05, 0.10, 0.20, 0.30, 0.40, 0.50\}$ with 15 scenarios per level, using the same random seed at each level. Patterns were assigned from the median e_{fit} at each level as described in Results. As in the selection, e_{fit} was computed to first order for the problems marked in Table 1. For Chen, between 7 and 13 of the 15 scenarios per level could be simulated; the remaining scenarios failed to integrate and were discarded.

Formal verification

The formal development uses Lean 4 (version 4.23.0) and Mathlib[17, 18]. Model regularity (smoothness of the vector field, existence of solutions on the simulated horizon and non-negativity of concentrations) was verified for 22 benchmark models by translating their SBML rate laws into Lean expressions and applying verified checkers. All files compile without `sorry`, and `#print axioms` reports only the standard axioms of Lean’s logic (`propext`, `Classical.choice`, `Quot.sound`).

Use of large language models

An AI assistant (Claude, Anthropic) was used to help write and debug analysis code and Lean proofs and to edit the manuscript text. All code, proofs and results were checked by the authors, who take full responsibility for the content.

429 **Data availability.** The PETA benchmark problems analysed in this study are
 430 publicly available in the Benchmark-Models-PETA collection ([https://github.com/](https://github.com/Benchmarking-Initiative/Benchmark-Models-PETA)
 431 [Benchmarking-Initiative/Benchmark-Models-PETA](https://github.com/Benchmarking-Initiative/Benchmark-Models-PETA)). The social cognitive theory
 432 model is provided as Python code in the repository below. All results generated in
 433 this study (selected subsets, validation, robustness and sensitivity analyses) are avail-
 434 able at <https://github.com/monicammr/archivos> and will be archived on Zenodo upon
 435 acceptance.

436 **Code availability.** The Python analysis code and the Lean 4 formal development
 437 are available at <https://github.com/monicammr/archivos> and will be archived on
 438 Zenodo upon acceptance.

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 442 method. M.M.R. wrote the software and the formal proofs, performed the analyses
 443 and wrote the first draft. C.S., C.M. and A.A. supervised the work, contributed to
 444 the validation of the results and revised the manuscript. All authors reviewed and
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Table 1 Per-problem results. p , number of estimated parameters; m , number of measurements; $|\Theta_C|$, calibration parameters (always re-estimated); $|S|$, selected biological parameters; Stage, stage that produced the subset; e_{rel} and e_{fit} , median relative error over 15 scenarios at $\pm 5\%$ without and with re-estimation (admissibility: $e_{\text{fit}} \leq 43.6\%$; italic, not admissible; *, e_{fit} computed to first order because of simulation cost; †, not defined because the model has no solution on the measured horizon for 1% parameter changes); Pattern, robustness pattern over perturbation sizes from $\pm 1\%$ to $\pm 50\%$ (see text); Test χ^2/n , mean squared noise-scaled residual on held-out experiments for the reduced and the full model, both re-estimated on the training experiments (values near 1 mean errors of the size of the measurement noise; –, not evaluable or not run).

System	p	m	$ \Theta_C $	$ S $	Stage	e_{rel} (%)	e_{fit} (%)	Pattern	Test χ^2/n reduced / full
Alkan	44	1733	10	2	1	20	12.1*	A	2.75 / 3.38
Armistead	14	58	0	2	2	13	11.7	B	1.22 / 1.37
Bachmann	113	541	74	2	2	53	33.7*	A	4.77 / –
Beer	72	27132	0	18	2	57	50.1*	–	–
Bertozzi	8	22	0	2	2	58	32.9	B	–
Blasi	9	252	0	2	1	41	40.2	B	–
Boehm	9	48	0	2	2	60	9.4	A	2.57 / 11.35
Borghans	23	111	2	2	2	51	4.3	A	0.74 / 0.74
Brannmark	22	43	3	2	2	67	7.8	A	5.79 / 2.15e+03
Bruno	13	77	0	5	1	46	37.8	A	0.91 / 7.86
Chen	155	120	3	2	2	53	8.2*	A	–
Crauste	12	21	0	1	2	–	<i>n.d.</i> †	–	–
Elowitz	21	58	2	2	1	65	35.8	B	1.50 / 4.45
Fiedler	22	72	8	2	2	58	14.7	A	11.21 / 14.11
Fujita	19	144	3	2	2	81	29.4	A	5.94 / 8.04
Giordano	50	313	0	2	2	108	5.8	A	<0.01 / 0.06
Isensee	46	687	11	2	1	52	7.6*	A	1.08 / 1.08
Lang	294	9600	16	2	2	94	16.1*	A	8.98 / –
Laske	13	42	2	2	1	14	10.1	A	0.76 / 1.88
Liu	9	100	0	2	2	47	26.1	A	0.95 / 1.02
Lucarelli	84	1755	0	10	2	60	43.5*	C	1.57 / –
Okuonghae	16	92	0	2	2	75	4.1	A	4.67e4 / 1.22e5
Oliveira	12	120	0	2	2	134	12.9	A	915 / 8.15e4
Perelson	3	16	0	2	1	0	0.0	A	<0.01 / <0.01
Rahman	9	23	0	2	2	70	3.7	A	<0.01 / 0.03
Raia	39	205	5	2	2	32	3.3	A	8.23 / 5.10
Raimundez	136	627	79	2	2	108	37.2*	A	1.19 / –
SalazarCavazos	6	18	0	2	2	41	18.7	A	102 / 124
Schwen	30	286	15	2	2	42	17.1	A	8.20 / 8.40
Smith	25	62	9	2	2	32	7.8	A	1.50e4 / 1.50e4
Sneyd	15	135	0	2	2	56	38.1	A	2.22 / 2.55
Weber	36	135	5	2	2	93	4.1	A	1.00 / 226
Zhao	28	82	0	3	2	70	29.6	A	0.35 / 0.35
Zheng	46	60	0	2	2	95	30.0	A	1.58 / 20.60

Table 2 Sensitivity of the selection to the method settings. The complete method was re-run on 26 problems changing one setting at a time (baseline: $\delta = 1\%$, perturbation $\pm 5\%$, $\tau_R = 0.89$, $\tau_\kappa = \tau_{\text{VIF}} = 10$, $\tau_e = 0.436$). Same S , number of problems in which the selected subset is identical to the baseline; $|\overline{S}|$, mean number of selected biological parameters among admissible problems.

Setting	Admissible	$ \overline{S} $	Same S
Baseline	25	2.16	26/26
$\delta = 0.1\%$ / $\delta = 5\%$	25 / 25	2.16	26/26 / 26/26
Tolerances $10^{-9}/10^{-12}$ / $10^{-6}/10^{-8}$	25 / 25	2.16	25/26 / 26/26
Perturbation $\pm 1\%$ / $\pm 10\%$	24 / 25	2.16	26/26 / 26/26
$\tau_R = 0.80$ / 0.85 / 0.95	25	2.16	26/26
$\tau_\kappa = \tau_{\text{VIF}} = 5$	25	2.16	23/26
$\tau_\kappa = \tau_{\text{VIF}} = 30$	25	2.16	22/26
$\tau_e = 0.10$	22	3.64	12/26
$\tau_e = 0.20$	24	2.71	17/26
$\tau_e = 0.30$	25	2.44	20/26
$\tau_e = 0.50$	25	2.08	24/26

Table 3 Formally verified statements (Lean 4/Mathlib). Each statement is proved without unproved assumptions or additional axioms. The corresponding formal theorem names are given in Supplementary Table S3.

Step of the method	Verified statement
Energy and R_{var}	$E_j = (J^\top J)_{jj}$; the total energy is the sum of retained and discarded energy; the first-order output change caused by discarded parameters is bounded by their energy
Sensitivities of the ODE	For C^1 vector fields, solutions depend differentiably on parameters and initial conditions, and the sensitivity solves the variational equation
Greedy selection	The returned subset satisfies $\kappa \leq \tau_\kappa$ and $\text{VIF} \leq \tau_{\text{VIF}}$; rejected candidates would still violate the test with the final subset
Identifiability	κ and VIF bounds imply uniqueness and stability of least-squares estimates of the retained parameters
Admissibility threshold	Theorem 1 (minimum error after rescaling equals $\sqrt{1 - \cos^2}$)
Re-estimation	Adding parameters to the re-estimated set cannot increase the fitting error; $e_{\text{fit}} \leq e_{\text{rel}}$
Backward elimination	Theorem 2 (minimality with respect to inclusion)
Measured outputs	Error bounds on states transfer to outputs $y_i = h(g_i(x))/\sigma_i$ with Lipschitz observables; $\log$ and $\log_{10}$ are Lipschitz on $[c, \infty)$
First-order e_{fit}	The linearised fitting error never exceeds the rescaled error and differs from the exact one by at most a second-order term
Model assumptions	For 22 of the benchmark models translated from SBML, the vector field is smooth, solutions exist on the simulated horizon and concentrations remain non-negative